Quantum informatics

Quantum computing

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https://www.qworld.net

https://qczech.org

Outline

IV. Quantum computing

- *How do quantum computers look like?*
- *Where does the power of quantum computers come from?*
- *Quantum supremacy, quantum advantage*
- *Near-term applications*

V. Programming quantum computers

- Qiskit
- jupyter

Download and install: shorturl.at/wAMV9

IV. Quantum computing

Classification

Quantum computers

Hardware platform

Superconducting
trapped ion/atom
NV centre
photonic

Architecture

Gate based
measurement based
continuous variable
annealing

Trapped ion

Quantum computer platforms: the hardware

QC's come in many sizes and favours

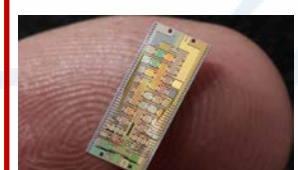
Support hardware
electronics, cooling, lasers ...

QPU
Quantum processing unit

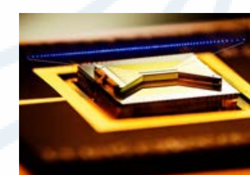
Part of ordinary computing infrastructure

Superconducting

Photonic



Trapped ion/atom



And more...!

Quantum computing architectures: program

Quantum annealing

D:WAVE
The Quantum Computing Company™

- tuneable quantum dynamics (Hamiltonian)
- continuous time
- discrete state space (qubits)

- Optimization problems

Ocean

Gate-based QC

IBM Quantum

Google IONQ

- set of quantum logic gates
- discrete time
- discrete state space (qubits)

- General purpose QC
- Sampling problems

Qiskit Cirq

Photonic QC

XANADU QUIX

中国科学技术大学
University of Science and Technology of China

QUANDELA

- set of quantum logic gates
- discrete time
- continuous space (qumodes)

- General purpose QC
- Sampling problems +

STRAWBERRY FIELDS

Source of power: how qubit are added

Single qubit: $|\psi\rangle \in \mathcal{H}_2$

Two qubits: $|\psi\rangle \in \mathcal{H}_2 \otimes \mathcal{H}_2$

Kronecker product

The basis by elementary tensor products:

$$|00\rangle = |0\rangle \otimes |0\rangle = |0\rangle|0\rangle$$
$$|01\rangle = |0\rangle \otimes |1\rangle = |0\rangle|1\rangle$$
$$|10\rangle = |1\rangle \otimes |0\rangle = |1\rangle|0\rangle$$
$$|11\rangle = |1\rangle \otimes |1\rangle = |1\rangle|1\rangle$$

"binary" notation
(0,1,2,3)

shorthand

Many qubits: high complexity

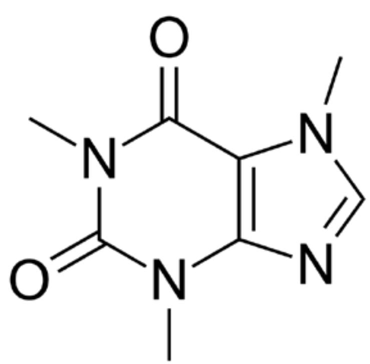
Quantum bits

- 2^n basis states
- $|\psi\rangle \in \mathbb{C}^{2^n}$

$$\begin{array}{ccc} 000\ldots000 & & |000\ldots000\rangle \\ 000\ldots001 & \rightarrow & |000\ldots001\rangle \\ & & \\ 111\ldots111 & & |111\ldots111\rangle \end{array}$$

Exponentially large linear space

Some important organic molecule



Caffeine

- 2^{48} bits
- Number of atoms in the universe

Quantum computing

Quantum systems are hard to compute!

Build quantum systems to do the job

Solve abstract/general problems?

Richard Feynman

David Deutsch

Peter Shor

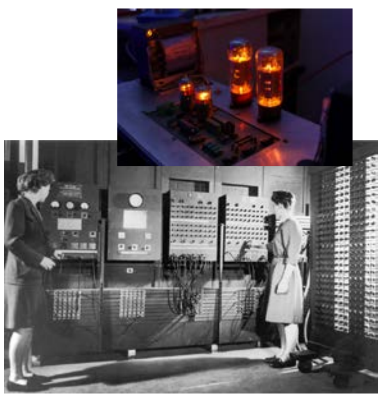
Scott Aaronson

Scheme of quantum computing

Hybrid quantum-classical computation

bits

qubits



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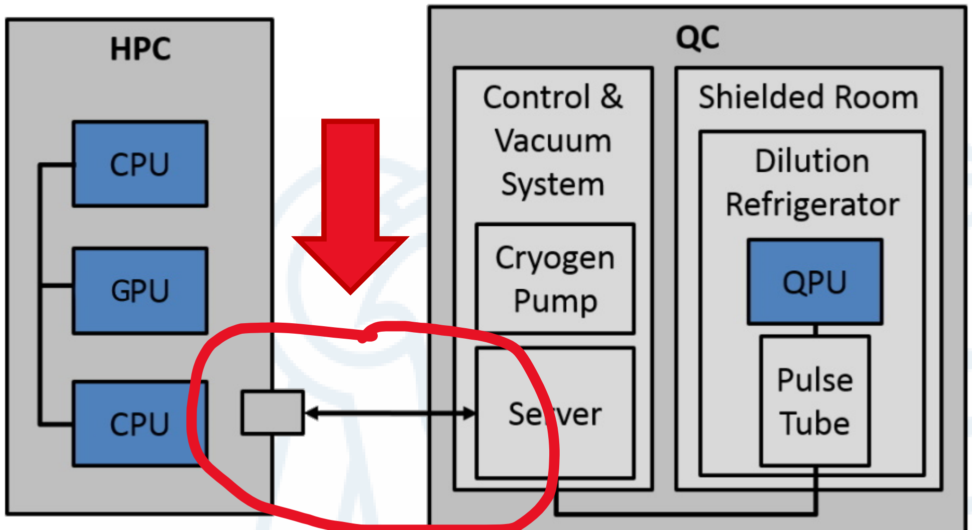
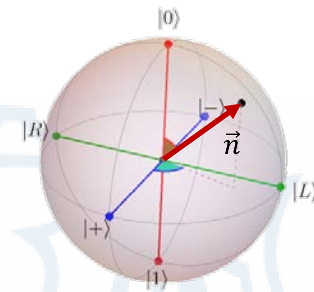
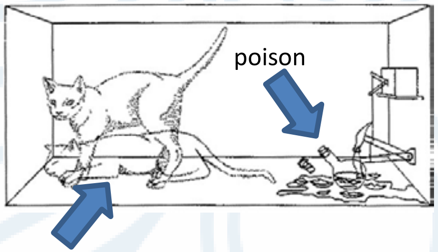






Schrödinger's cat

What can quantum computers do and not do?

Cannot

- speed up all computations

Not just a faster way of doing computations!

Can

- outperform ordinary computers solving carefully crafted problems
- execute small-scale algorithms solving real-world problems

No speedup in solving practical problems

Expected ⟹

Short and mid-term

- **optimization:**
  logistics, train schedule
- **machine learning:**
  (in science and IT)
- **quantum system calculations:**
  material science, chemistry, medicinel, quantum technology design

Long term

- cybersecurity
- …. *whole new industry*

How are quantum computers faster?

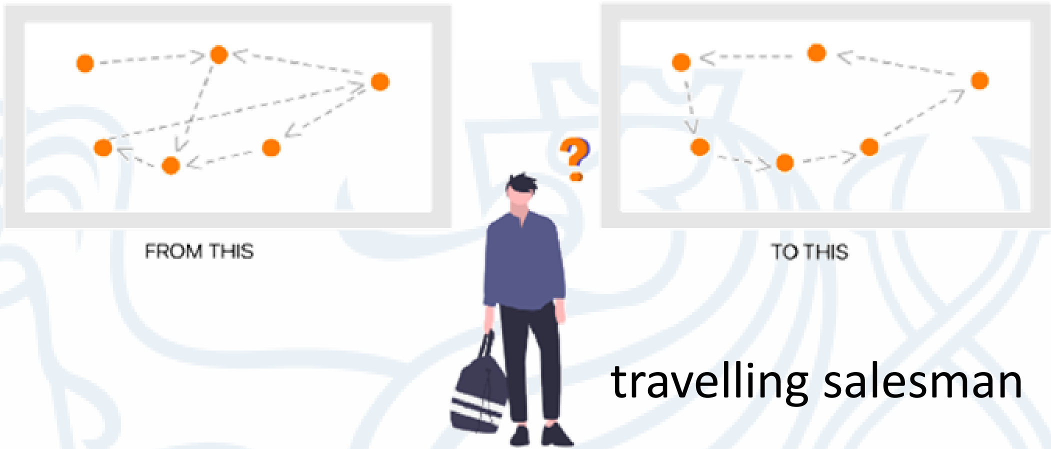
„Easy" and „hard" problems

„Easy" problem: addition

$$\begin{array}{r} 54 \\ +45 \\ \hline 99 \end{array} \qquad \begin{array}{r} 5489254 \\ +3489345 \\ \hline 8978599 \end{array}$$

No. digits	operations
2	2
7	8

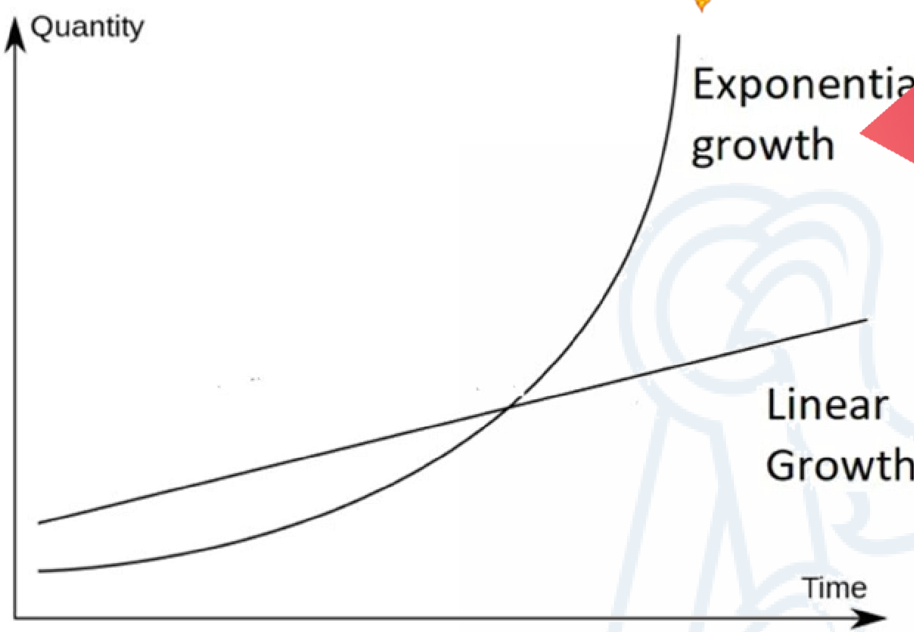
„Hard" problem



travelling salesman

How are quantum computers faster?

„Easy” and „hard” problems



„Hard”
$\exp(n)$

„Easy”
$poly(n)$

Factorisation: exponential speedup

Peter Shor

Given integer N ($L \coloneqq \log_2 N$ -- the number of bits)

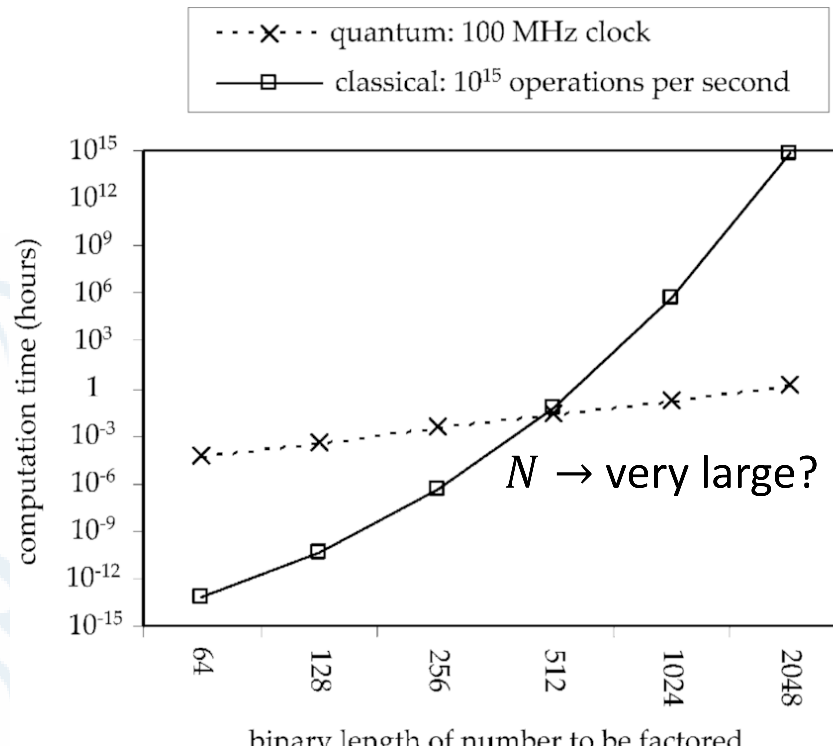
Find p such that p divides N

Difficult problem (NP)

“asymptotic scaling”

Classical: $\sim$exponential

Quantum: $\sim$ polynomial



New complexity class: BQP

bounded-error quantum polynomial time

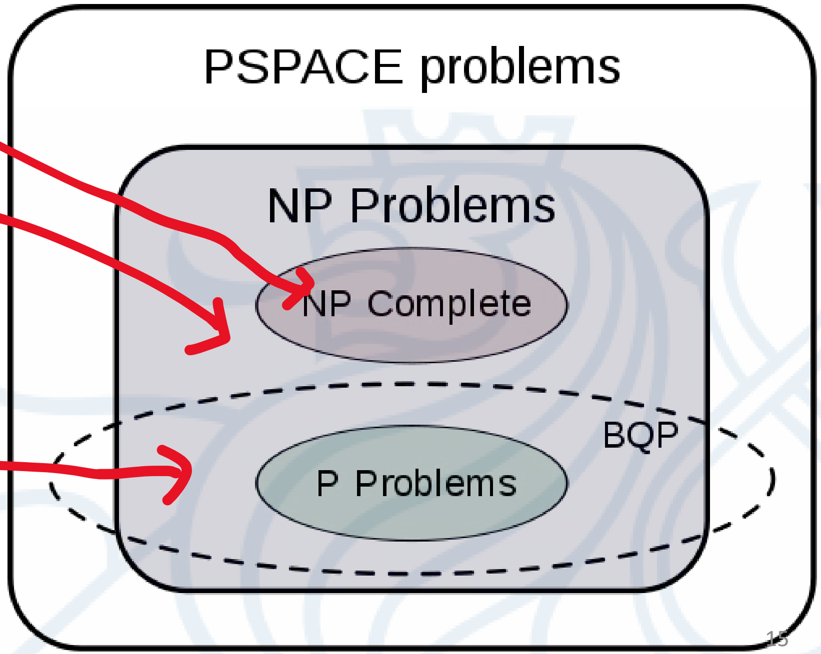
Not known if **NP⊂BQP**

- travelling salesman problem
- …
- constructing time table for a school

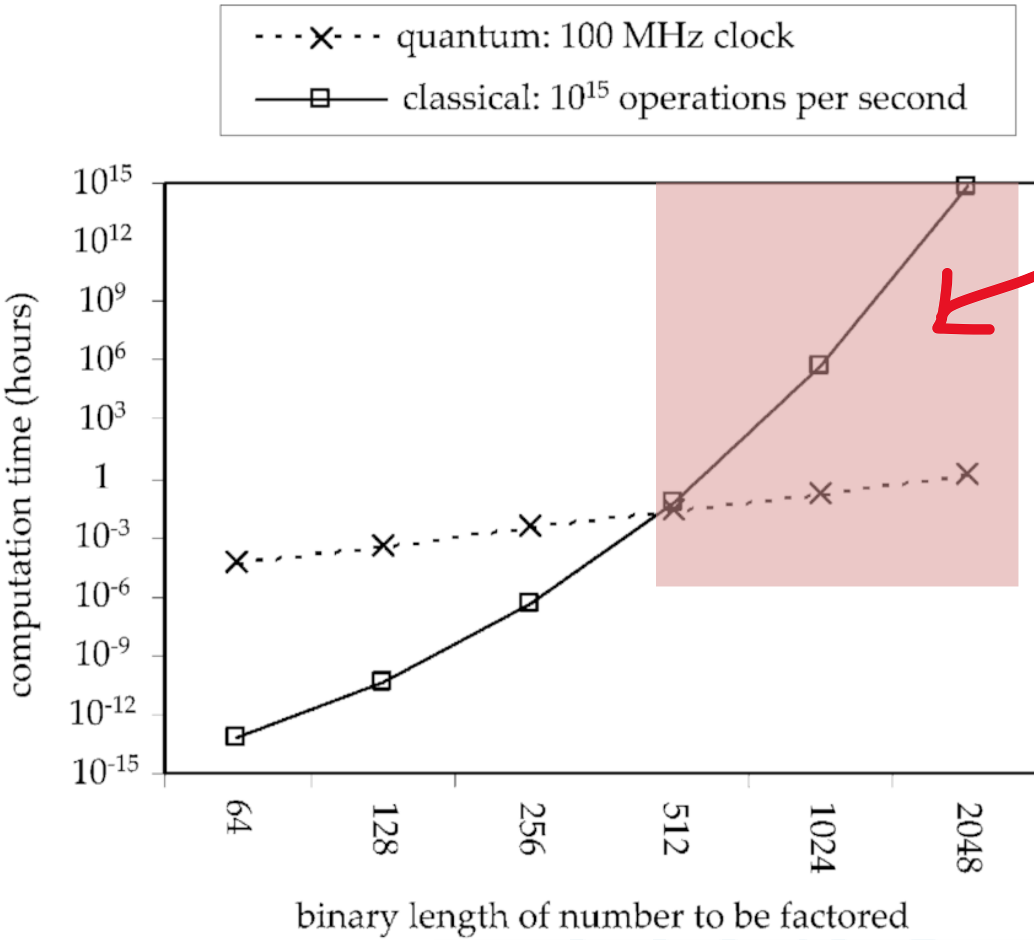
- Some **NP** problems in polynomial time

Shor's algorithm

(hacking RSA encryption)



When are quantum computers faster?



Quantum supremacy

No practical problems are here

prime factors

$15 = a \cdot b$

$79 = a \cdot b$

$86 = a \cdot b$

$828263 = 853 \cdot 971$

Quantum supremacy demonstrations

Sampling

- Random circuits or non-classical states of light
- Generate a random variable from a given distr.

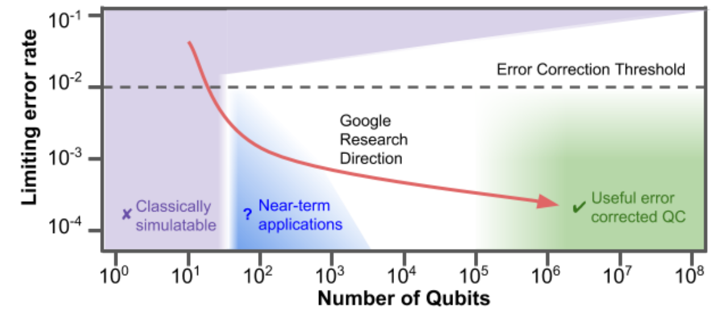
Classically hard, quantumly easy

An experiment that is hard to simulate classically

- No obvious practical use
- Ion trap QC (many degrees of freedom)

Strengths

- Beats presently known classical solutions
- Complexity in the controlled degrees of freedom



Google Quantum AI

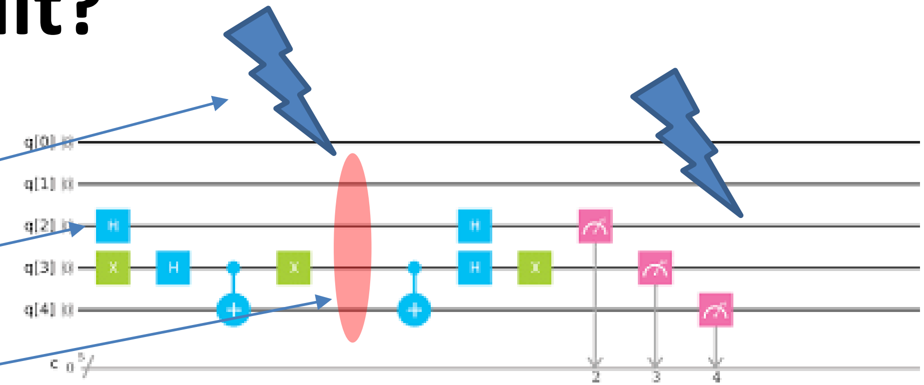
中国科学技术大学
University of Science and Technology of China

Questions

- Limited fidelity can affect simulability

What makes it difficult?

Errors
- Environment
- Accuracy of gates
- Cross – talk between qubits



Error correction
- Multiple copies?
- No cloning:

$$U|\psi\rangle|\phi\rangle = |\psi\rangle|\psi\rangle$$

no such unitary op.

- Additional qubits necessary!

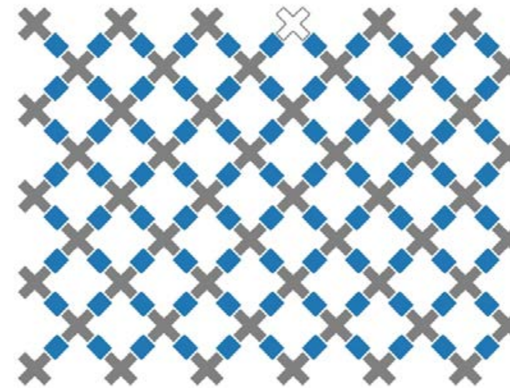
Logical qubits
- Physical qubits have errors
- Join multiple physical qubits + error correction = 1 logical qubit

Fault-tolerant quantum computing

State of the industry in HW

Google

Sycamore: 53 qubits -> 72



IBM **Quantum**

IBM Q: 433 qubit (Osprey) -> 1024 qubits

smaller: free to use via web interface

中国科学技术大学

University of Science and Technology of China

UCTS:

- 66 qubits (superconducting)
- 70 qumodes (optical)

Quantum supremacy

NISQ:
Noisy intermediate scale quantum computing

- Only physical qubits (errors present)
- Up to 433 qubit capacity
- Power: demonstrated ☑
- Practical advantage: not demonstrated ✗

NISQ algorithms

- Shallow circuits + classical processing
- Optimization
- Variational methods:
  - finance, quantum chemistry
- Machine learning (QNN)
- Simulation of quantum systems

Doesn't have to be precise:
just better than classical

Beyond the NISQ era?

Factoring (error correction needed)
- keys of several thousands of bits
- Need: Millions of physical qubits!

Maybe in the next 10 years?

Types of quantum algorithms

Exact

Numerical pr

- Find … that $\frac{N}{p} = q$
- $(N, q$

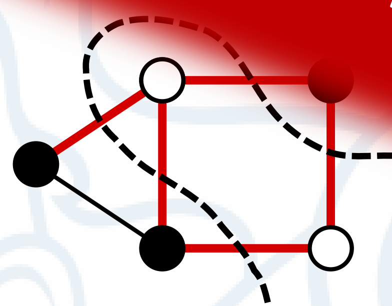
Shor's algor

Only error-free quantum computers

Graph property

- How to cut graphs in half?
- Are two graphs the same?

Grover's algorithm



Types of quantum algorithms

Approximate numerical

Optimization:

- ”More optimal” than classical is an advantage!

Properties of quantum systems:

- Approximation: e.g. chemical reaction energy

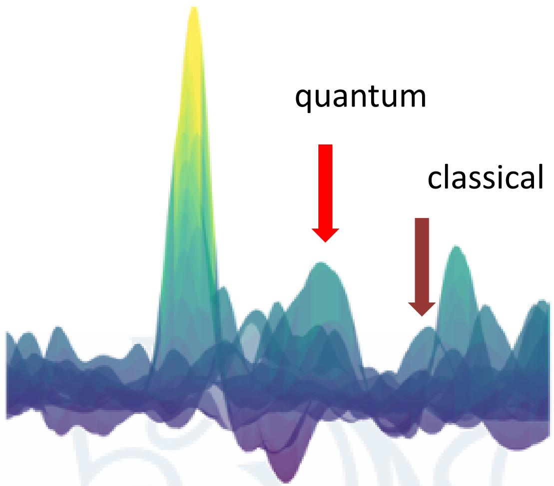
Probabilistic

Machine learning:

- Train to provide useful answer with high chance

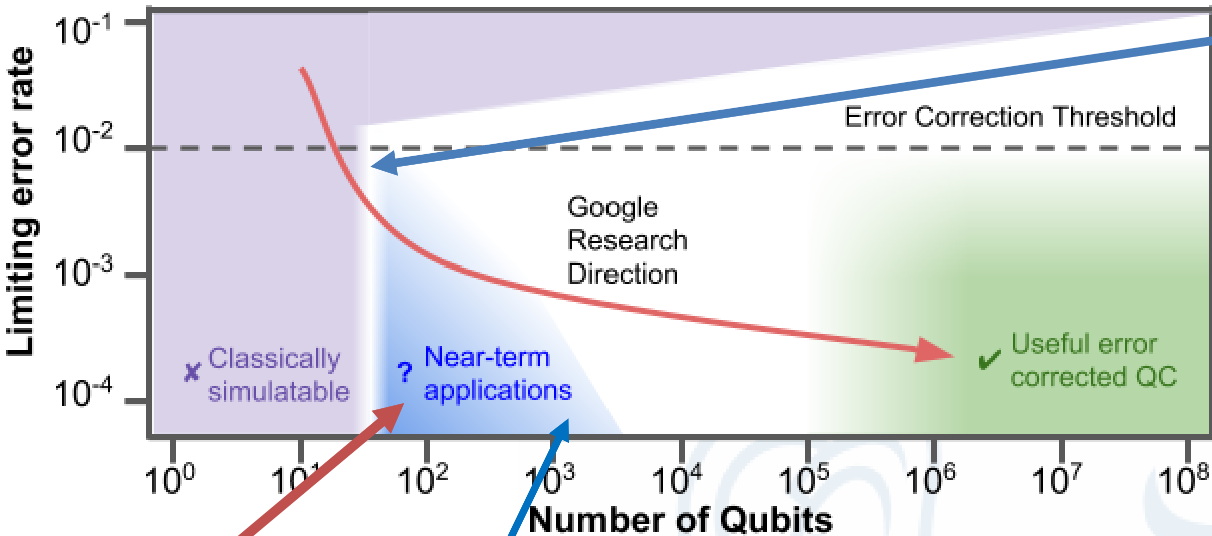
Sampling problems:

- Produce answer w/ pre-scribed distribution



Quantum supremacy

Google’s roadmap (2018)



Google probably got here (October 2019)

NISQ regime

IBM 433 qubit processor

Quantum supremacy

QC proves superior to any classical HPC

Quantum advantage

practical benefit from QC

Programming quantum computers

How to program quantum computers?

- Jupyter notebook
- qiskit

Create small quantum circuits

- See the coding principles
- experience quantum features

Notebooks:

Notebooks will be shared after lecture

Quantum circuit model

what happens to the initial vector?

$$\left|\psi_{final}\right\rangle = U\ |\psi_{initial}\rangle$$

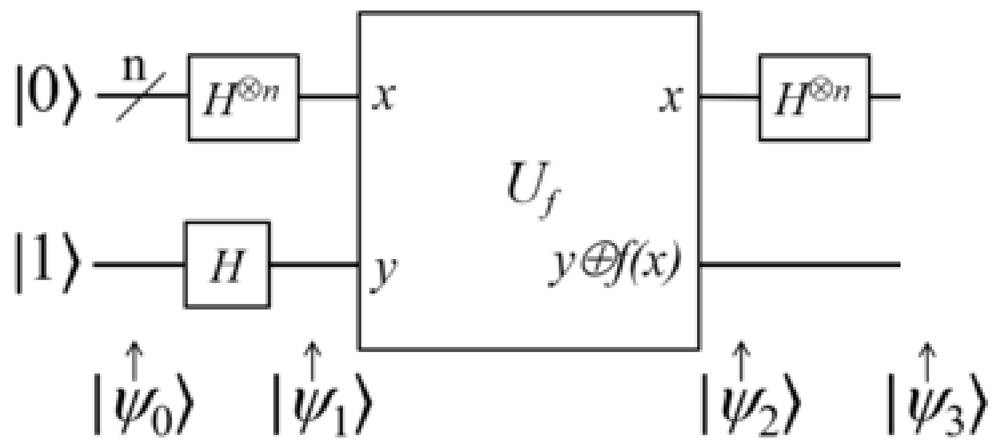
- $\left|\psi_{final}\right\rangle, |\psi_{initial}\rangle \in \mathbb{C}^4$
- U unitary $\mathbb{C}^4 \rightarrow \mathbb{C}^4$

input

output

qubit 1

qubit 2

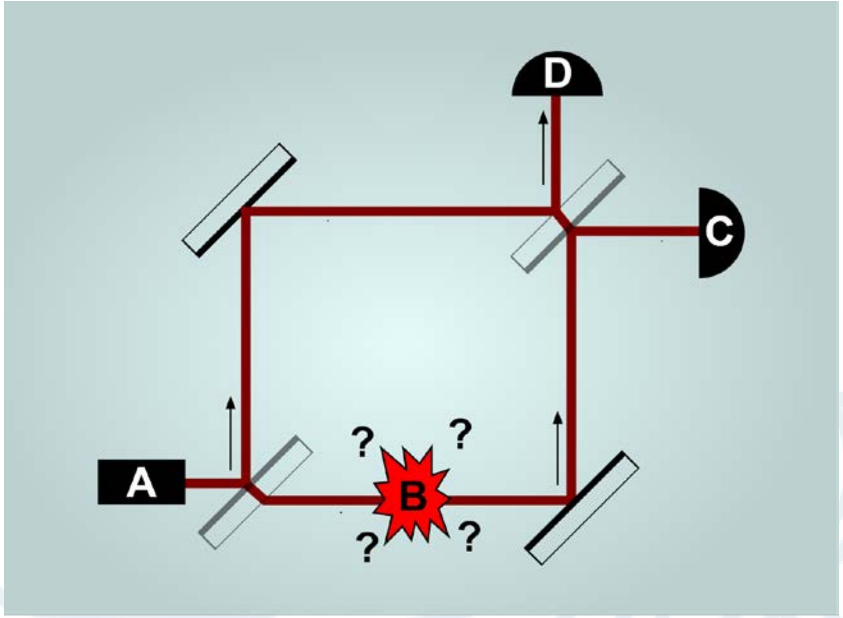


Probability vs. amplitude

With probability vectors

- $\mathcal{P}(0) = (1,0)$
- $\mathcal{P}(1) = (T, R)$
- $\mathcal{P}(2) = (TT + RT, TR + RT)$

$$T + R = 1$$



Amplitudes

$$|u|^2 + |v|^2 = 1$$

- $\psi(0) = (1,0)$
- $\psi(1) = (u, v)$
- $\psi(2) = (uu^* - rr^*, ru^* + ur^*)$

"B"

$$r = \frac{i}{\sqrt{2}} \to r^* = -r$$

$$\psi(2) = (1,0)$$

Quantum bit

Amplitudes

$$\psi = (\alpha, \beta)$$

α, β complex numbers
$|\alpha|^2 + |\beta|^2 = 1$

Basis states in $\mathbb{C}^2$

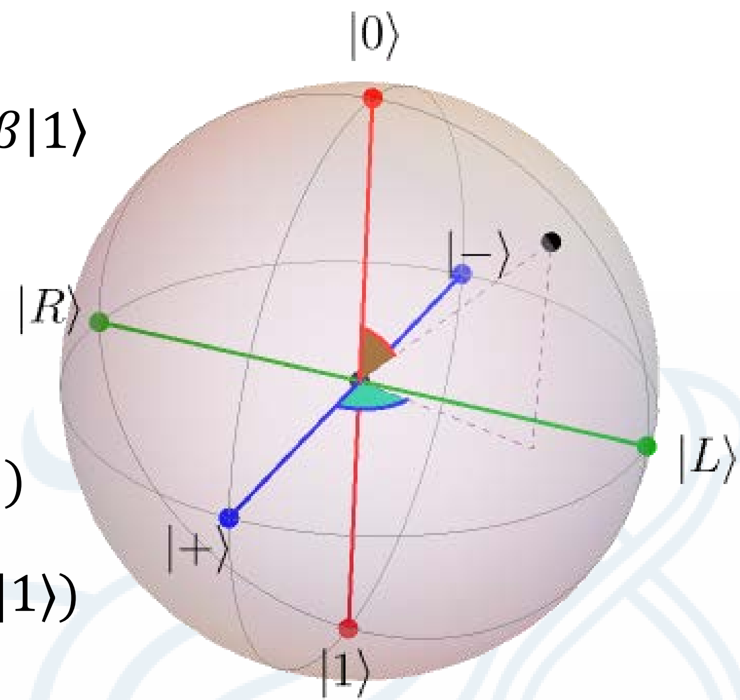
$$|0\rangle = (1,0)$$

$$|1\rangle = (0,1)$$

Dirac notation

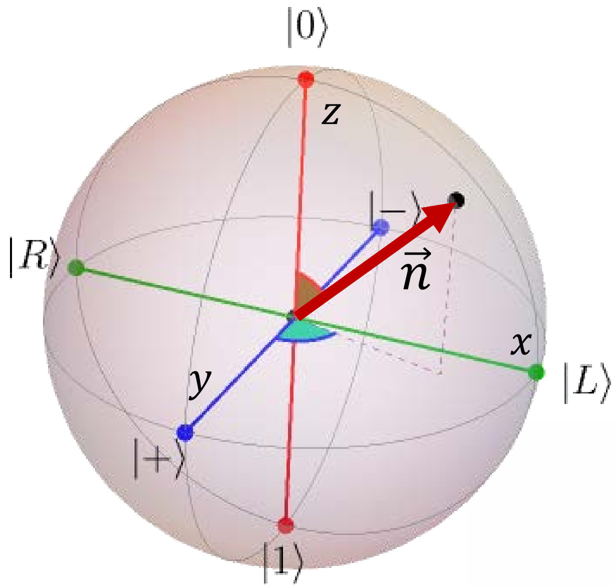
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$|\pm\rangle = \frac{1}{\sqrt{2}}(1, \pm 1)$$
$$= \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$$



https://javafxpert.github.io/grok-bloch/

State of qubit: Bloch sphere



$$|\psi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle$$
$$= \cos\theta/2\,|0\rangle + e^{i\varphi}\sin\theta/2\,|1\rangle$$

point on the surface of sphere

Spherical coordinates: $\vec{n} = (r, \theta, \varphi)$

$\theta \in [0, \pi],\ \varphi \in [0, 2\pi], r = 1$

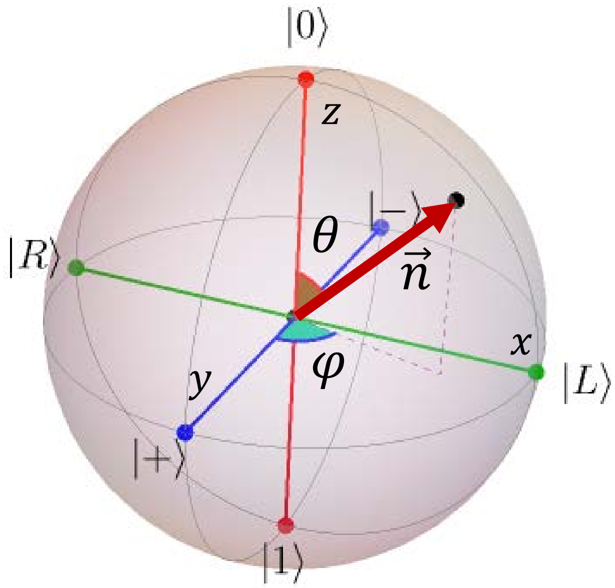
$$|\pm\rangle = \frac{|0\rangle \pm |1\rangle}{\sqrt{2}}$$
$$|R/L\rangle = \frac{|0\rangle \pm i|1\rangle}{\sqrt{2}}$$

We have used: $|\alpha|^2 + |\beta|^2 = 1$ "normalization"

$\alpha \in \mathbb{R}$ "global phase"

https://javafxpert.github.io/grok-bloch/

Operations



$$|\psi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle$$

point on the surface of sphere

Operations: unitary

$$U^\dagger U = \mathbb{I}$$

Rotation of Bloch vector

$$SU(2)$$

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

“computational basis”

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$U = \begin{pmatrix} u & v \\ -v^* & u^* \end{pmatrix}$$

$$|u|^2 + |v|^2 = 1$$

https://javafxpert.github.io/grok-bloch/

Measurement: “Born rule”

Computational basis: $|0\rangle, |1\rangle$

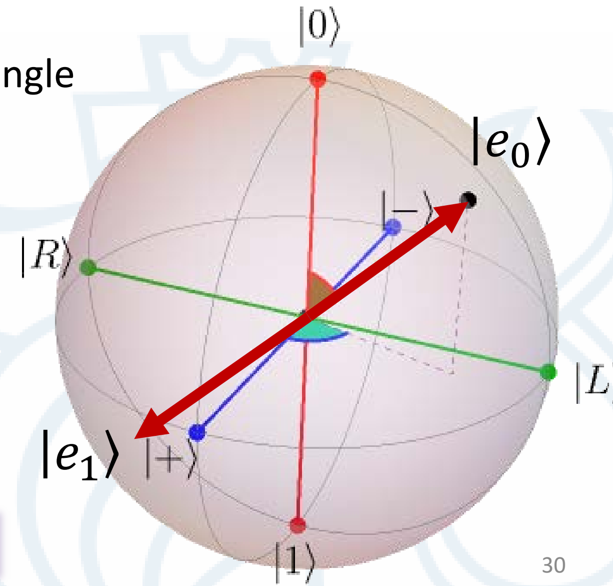
$$|\psi\rangle = \cos\theta/2\,|0\rangle + e^{i\varphi}\sin\theta/2\,|1\rangle = a|0\rangle + b|1\rangle$$

Outcome probability from amplitudes:

“0”: $|\langle 0|\psi\rangle|^2 = |\cos\theta/2|^2$

“1”: $|\langle 1|\psi\rangle|^2 = |\sin\theta/2|^2$

Δ: enclosed angle



Two orthogonal states: $\langle e_0|e_1\rangle = 0$

$|e_0\rangle, |e_1\rangle$

axis on the Bloch sphere

Programming quantum computers

How to program quantum computers?

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Quantum supremacy background

#P: counting problems associated with the decision problems in the set NP

- $\mathrm{f}(x): \{0,1\} \rightarrow \{0,1\}^n$
- $\mathrm{g}(x) = \sum f(x)$
- E.g. Permanent of a matrix
- Random classical circuits

GapP

- number of accepting paths of M minus the number of rejecting paths of M
- $\mathrm{f}(x): \{0,1\} \rightarrow \{-1,1\}^n$
- $\mathrm{g}(x) = \sum f(x)$
- Random quantum circuits

Approximating GapP is hard